

Blackout

AP Physics 2 · circuits & induction — when the course introduces a concept, against where the story is standing when it does.

Calder Switching Station, and two circuits of lattice towers walking off the edge of the map.

45	25/32	0.29	0	34	L3 of 6
STOPS · 15 DAYS	CONCEPTS CLAIMED	DAY ↔ CONCEPT (P)	PREREQUISITE INVERSIONS	DECLARED AS READ	DAY 1 OPENS AT

READING Where this campaign stands

Both slates are applied, and the rule is a check now. Before them, day 1's first stop asked what a falling frequency trend is evidence of and the card named c20 · inertia, governor response and droop control (twice) and c31 · metering — layer 5 of 6 — while c19 · frequency as the balance of supply and demand, which droop is built out of, was claimed by no stop until day 12, and c18 · synchronous machines by no stop at all. Today that stop claims the balance itself, which is what it computes, and says on the card that it takes synchronous machines and power-against-energy as read. engine/dev/conceptOrder.mjs is green on all 29 themes.

The numbers: of 28 prerequisite inversions, 12 rested on material this campaign itself teaches later and 16 on the bottom of the graph. The first group is ordered — 0 left — and the second is declared, in 32 takesAsRead entries that the importer refuses unless the concept is genuinely something the stop's claim is built out of. Concepts claimed went 22 → 25 of 32; p went 0.19 → 0.29.

The gate is per stop, and that is not the same audit as this document's. Two stops can claim one concept, and each of them has to answer for its own prerequisites — so conceptOrder found 6 rows the concept-level count had hidden, including a stop standing beside its own base on day 6 rather than after it. Four were declared; two were ordered, and one of those is the fix worth copying: day 10's dark-hours estimate now claims energy from power over time, which is what E = Pt is, and that single re-claim put the base under two later stops at once.

THE FINDING THAT DECIDES THE WORK

Re-ordering the story is not the fix. The fortnight is a chronology — the trip causes the aftermath, the island precedes the black start, the report is last — and searching permutations of the fifteen missions that respect those 11 constraints, the best correlation available is p = 0.46 against today's 0.29 — and measured before the pass it left 10 of the 12 in-course inversions standing, which is why it was never the fix. The re-order that was taken scores 0.29 rather than that ceiling on purpose: the search ranked orders by prerequisite breaks first and correlation second, and the correlation-maximal order teaches backwards. Ignore the story entirely and it reaches 0.50; move individual stops rather than whole missions and it reaches 0.90, at 30 rewritten scenes.

Correlation is the wrong target anyway — it is the cheap number adjacent to the thing. The target is the prerequisite: every concept a stop claims has a base that arrived on an earlier day, or the stop says out loud that it assumes it. That is what the two slates did, without moving a mission or touching the story: 12 in-course inversions down to 0 — two swapped pairs, one rewritten question, and the last row declared rather than moved, because day 1 cannot have anything before it.

DEPENDENCY What this course is built out of

Authored beside the concepts in `tools/syllabus.js` — where `needs` already lived for equations — because it is a claim about the subject and not about this game, and `concept0rder.mjs` reads it there. It is not difficulty — `c20` is no harder than `c9`, it is further downstream — and it is not the syllabus's own order, which lists transformers at 13 and Faraday's law at 17. Argue with this first: a wrong prerequisite here produces a confident wrong number everywhere else in the document.

LAYER LICENSED FROM CONCEPTS

L0	d1	c1 Charge, current, voltage and resistance
L1	d1	c2 Ohm's law and resistive networks · c6 AC waveforms: frequency, period and phase
L2	d4	c3 Kirchhoff's laws and nodal analysis · c5 Electrical power and energy over time · c7 RMS against peak and average · c8 Capacitance, inductance and stored energy · c17 Faraday's law and electromagnetic induction · c23 Protection: relays, breakers and coordination
L3	d7	c4 Series, parallel and equivalent circuits · c9 Reactance, impedance and phasors · c13 Transformers, turns ratio and losses · c15 Transmission losses and conductor sizing · c18 Synchronous machines and synchronisation · c21 Load curves and demand forecasting · c29 Storage, round-trip efficiency and ramp rate · c31 Metering, instrument transformers and measurement error
L4	d9	c10 Real, reactive and apparent power · c14 Voltage levels and the per-unit system · c19 Frequency as the balance of supply and demand · c22 Economic dispatch and merit order · c24 Fault current and short-circuit analysis · c27 Contingency analysis and the N-1 criterion
L5	d12	c11 Power factor and its correction · c12 Three-phase systems and balance · c16 Voltage drop, regulation and reactive support · c20 Inertia, governor response and droop control · c25 Earthing, step-and-touch potential and safety · c28 Cascading failure and the sequence of events
L6	d15	c26 Islanding, black start and restoration order · c30 Renewables: variability, curtailment and inverters · c32 Load shedding as a decision under uncertainty

Licensed from is a report and must never become a gate: it assumes a course spreads its layers evenly over its days, which no course promises. On that report 20 of 45 stops sit before their layer opens, and six of those are the black-start block — layer 6 material, correctly late in the course, landing on days 9–11. A gate on this number would have sent the work at six stops that are right.

AUDIT Every day, and the layer it teaches at

d1 L≤1 c29·L3 c13·L3 c3·L2	d2 L≤1 — c15·L3 —	d3 L≤1 c19·L4 c19·L4 c31·L3	d4 L≤2 c23·L2 c2·L1 c9·L3	d5 L≤2 c4·L3 c15·L3 —	d6 L≤2 c10·L4 c18·L3 c7·L2	d7 L≤3 c4·L3 c29·L3 c16·L5	d8 L≤3 c24·L4 c17·L2 c21·L3	d9 L≤4 c20·L5 c8·L2 c26·L6	d10 L≤4 c5·L2 c26·L6 c26·L6	d11 L≤4 c31·L3 c15·L3 c26·L6	d12 L≤5 c27·L4 c5·L2 c19·L4	d13 L≤5 c25·L5 c22·L4 c22·L4	d14 L≤5 c15·L3 c30·L6 c27·L4	d15 L≤6 c29·L3 c5·L2 —
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Each column is a day: the layer the report licenses by then, and the layer each stop actually claims. Red is above the licence. What matters is not the red — it is that there is no slope at all. Days 1 and 2 teach layers 3, 3, 2, 3; days 12 to 15 teach 4, 2, 4, 5, 4, 4, 3, 6, 4, 3, 2.

WORK Nothing is out of order

Every claim's prerequisites arrive on an earlier day or are declared on the card — 34 declarations across 22 stops. What follows is the record of the twelve that were out of order, because the shape of each fix is the useful part for the next game: two were a claim naming the wrong concept, three were a stop standing beside its own base, and two needed a question the course did not have.

The 12, and what fixed each

THE ROWS, AS THEY STOOD	THE FIX	COST
c20 d1 ← c19 d3	Day 1's two stops claim c19 , which is what they already compute: one number for the whole interconnection, and df/dt as that balance's rate. Droop is claimed on day 9 anyway.	claim
c26 d9 ← c18, c19	Synchronising is what day 4's breaker-closing SEQUENCE teaches; claim it there. With c19 on day 1, the whole black-start block is legal where it stands.	claim
c16 d3 ← c9, c10, c15	Voltage drop rests on impedance and on real against apparent power, and neither is mentioned anywhere in the fortnight — the same hole the conversion plan records as its one real equation gap, $\Delta V \approx I(R \cos \phi + X \sin \phi)$. Two questions get written, on days 2 and 3, and voltage drop moves to day 8.	2 questions
c12 d4 ← c10	Three-phase power from line quantities is $P = \sqrt{3} \cdot V \cdot I \cdot \cos \phi$, and the $\cos \phi$ is what day 3's new stop is about. Covered by the same question.	covered
c22 d8 ← c21 d8	The load curve moves from day 8 to day 7. One day, and its scene names no event.	1 day
c25 d7 ← c24 d7	Fault current moves from day 7 to day 6, ahead of the earthing stop that rests on it; series and parallel moves from day 6 to day 5, ahead of that.	2 days
c27 d2 ← c4 d6, c15 d3	Day 2's contingency stop becomes the impedance question; N-1 is claimed on days 12 and 14 as well, where both of its bases already exist.	covered

SLATES Three appetites

No slate adds a stop. Every figure in the three boxes is computed by applying the slate to the real content and re-running the audit.

A · Declare what it takes as read

DONE 18 AUG 2026 · 22 STOPS · NOTHING REWRITTEN

The 16 silent assumptions are authored now: `takesAsRead` on the stop, refused by the importer unless the concept is something the stop's claim is built out of, and printed to the player as an `assumes` line and on the Key concept door — "rests on ... (taken as read here)". It fixed no ordering, and it was never meant to. What it fixed is that the campaign leaned on pieces of prior physics without saying so, and a reader could not tell a deliberate assumption from an accident. It also has to come first: day 1 could otherwise only ever claim a concept with no prerequisites at all.

undclared assumptions **16 → 0** · declarations **32** across **22** stops · nothing moved, nothing was rewritten

B · Fix the twelve

DONE 18 AUG 2026 · RE-RUN THIS FILE TO RE-MEASURE

The 10 authored claims, the two swaps and the one rewritten question in the table above. No mission was re-ordered and no scene was rewritten except the one whose question changed. The figures on the right are the pass, measured before and after on the real content — everything else in this document already reads the game as it now is.

in-course inversions **12 → 0** · concepts claimed **22 → 25** · ρ **0.19 → 0.29** · declared assumptions **16 → 0**, because four more concepts claimed is four more sets of bases to declare · the row left over is day 1 itself, which no ordering can fix

The missions are in a new order and no stop was rewritten to get there: $d1 \leftarrow 5$ $d2 \leftarrow 13$ $d3 \leftarrow 1$ $d4 \leftarrow 2$ $d5 \leftarrow 3$ $d6 \leftarrow 4$ $d7 \leftarrow 6$ $d8 \leftarrow 7$ $d9 \leftarrow 9$ $d10 \leftarrow 10$ $d11 \leftarrow 11$ $d12 \leftarrow 12$ $d13 \leftarrow 8$ $d14 \leftarrow 14$ $d15 \leftarrow 15$. It was chosen by search over every order the chronology allows — 11 constraints read off the stakes — ranked by prerequisite breaks first and correlation second, because a re-order that reads well and teaches backwards is not an improvement. The fortnight opens on the reserve day and the three-problems day; the trip lands on **day 3**, which is what the swing equation being the first arithmetic anybody met was costing. Eight continuity edits (the weekday stamps, one back-reference to an event that has not happened yet, one scene tail) and one character introduced at her new first mention — `checkNames` caught that, having been told nothing about the re-order.

the swing equation first appears on **day 3**, not day 1 · p **0.24** → **0.29** · declarations **32** → **35** · prerequisite inversions **0** · no stop rewritten, no objective lost

GATE engine/dev/conceptOrder.mjs

One rule, and it is the rule `equationOrder` already enforces one field over: **for every stop that claims concept *c* on day *d*, each of *c*'s prerequisites is claimed by a stop on a day earlier than *d*, or is named in that stop's own assumes**. The hatch is what makes the rule satisfiable, and it is the right hatch — `assumes` already exists, is already printed, and turns "the player has not met this" into an authored sentence. Both halves get counted: the failures, and the exceptions, because a campaign with sixty exceptions has not passed anything.

Two reports beside it and neither of them a gate: the rank correlation, and the layer band. The selftest is the load-bearing half — three cases, each of them a bug this file could have. A fixture whose prerequisite is claimed on the **same** day must fail, because the engine opens a day in any order. One claimed a day earlier must pass. One declared in `assumes` must pass, with the exception counted and printed. Put each bug back and that case, and only that case, should fail.

Debt rather than an advisory flag: 0 rows on one game is too many to gate on cold, so what exists today goes in `concept-debt.json` beside `curriculum-debt.json`, with the same two properties — a row not on the list fails immediately, and a row on the list that has since been fixed **also** fails, naming the line to delete. It only shrinks.

THE ORDER OF WORK

1. Author the claim: `concept:` on a stop in the book, **refused** by `import-book.mjs` when it names something not on the theme's syllabus rather than dropped, with `pickKeyConcept` as the fallback. Nothing here can be built before this — a gate on a derived field grades the matcher, not the course.
2. Review the 32 needs lists into `tools/syllabus.js`; the graph in this document is one reading of one syllabus.
3. `conceptOrder.mjs` with its three selftest cases, and `concept-debt.json` baselined at today's 0.
4. `node engine/dev/curriculumDelivery.mjs blackout --snapshot before.json`.
5. Slates A then B, editing the book and re-importing — `node tools/import-book.mjs books/blackout.yml blackout --verify`. The two written questions are a deliberate curriculum change: re-snapshot after them and say so in the commit.
6. `--against before.json`, knowing what it will and will not say — a corrected claim is invisible to it, a reworded takeaway fails it, and a changed day is not compared at all, so add the day comparison to `diffSnapshots` as a **note** before slate C, never as a failure.
7. `npm run check blackout`, then `npm run drive blackout` on the two new questions, right answer and wrong.
8. Re-derive the editions. `blackout_fable` shares this syllabus and has the same day-1 opening; `blackout_ms` claims **no** concepts at all, because none of its 16 concepts has a `t` written — its cards show no key-concept door, and this whole audit is unmeasurable there until they do.

person should argue with. The conversion plan — format mix, equations owed, fun formats — is `plans/blackout.html`, and is a different question. Run the game with `THEME=blackout npm run dev`.

APPENDIX All 45 stops, in campaign order

#	DAY	AREA	FORMAT	CONCEPT	LAYER	TITLE
1	d1	LOAD	BALLPARK	c29	L3	What comes out against what went in
2	d1	GEN	CHOICE	c13	L3	The reason the wires are not at wall voltage
3	d1	OPS	CHOICE	c3	L2	What a node has to add up to
4	d2	OPS	DELEGATE	—	—	Which of the three cannot wait
5	d2	TRANS	BALLPARK	c15	L3	Loss when the load turns up anyway
6	d2	LOAD	SEQUENCE	—	—	An order that survives being carried out
7	d3	OPS	PROTOCOL	c19	L4	What a frequency trend is evidence of
8	d3	GEN	BALLPARK	c19	L4	How much power actually left the system
9	d3	METER	TRACE	c31	L3	The order the records claim things happened in
10	d4	TRANS	TRIGGER	c23	L2	What you do before a relay does it for you
11	d4	LOAD	BALLPARK	c2	L1	The power that never arrives
12	d4	OPS	BALLPARK	c9	L3	What the path opposes at 50 hertz
13	d5	DIST	PROTOCOL	c4	L3	What is common and what divides
14	d5	TRANS	PROTOCOL	c15	L3	What actually limits a conductor
15	d5	OPS	CHOICE	—	—	The order somebody will read back to you
16	d6	METER	BALLPARK	c10	L4	What three phases actually add up to
17	d6	GEN	SEQUENCE	c18	L3	Four things that have to agree before the breaker closes
18	d6	TRANS	CHOICE	c7	L2	Two instruments, two quantities
19	d7	TRANS	BALLPARK	c4	L3	How the flow splits between two paths
20	d7	OPS	VALUE	c29	L3	What is worth knowing before the peak
21	d7	DIST	BALLPARK	c16	L5	Why the far end sits low
22	d8	TRANS	BALLPARK	c24	L4	What a short circuit is limited by
23	d8	METER	CONTROL	c17	L2	A reading taken without a connection
24	d8	LOAD	BALLPARK	c21	L3	The peak, and how far the cold moves it
25	d9	OPS	DEGENERACY	c20	L5	Why the island's frequency is falling slowly
26	d9	METER	PROTOCOL	c8	L2	What is stored, and for how long
27	d9	DIST	CHOICE	c26	L6	Keeping it, or collapsing it on purpose
28	d10	GEN	BALLPARK	c5	L2	What the dark hours actually cost
29	d10	OPS	CHAIN	c26	L6	The order physics allows
30	d10	DIST	CASEBOOK	c26	L6	Why it cannot go faster
31	d11	METER	CHOICE	c31	L3	The sensor that was confident and wrong
32	d11	TRANS	SWEEP	c15	L3	What a rating is a statement about
33	d11	OPS	CASEBOOK	c26	L6	What survives a bad number
34	d12	OPS	CHOICE	c27	L4	The failure that has not happened
35	d12	LOAD	PROTOCOL	c5	L2	The trend with nothing wrong with it

#	DAY	AREA	FORMAT	CONCEPT	LAYER	TITLE
36	d12	GEN	VALUE	c19	L4	Which of these is a measurement
37	d13	DIST	CHOICE	c25	L5	The ground is not at one voltage
38	d13	OPS	CASEBOOK	c22	L4	Cheapest first, and the exceptions that are not exceptions
39	d13	GEN	SCIENCETANK	c22	L4	When the cheapest power cannot get out
40	d14	METER	CHOICE	c15	L3	Hours over limit, added up
41	d14	TRANS	SCIENCETANK	c30	L6	What the window is worth
42	d14	OPS	PROTOCOL	c27	L4	The 18 months you just bought
43	d15	METER	ATTEST	c29	L3	Measured, inferred, assumed
44	d15	LOAD	CHOICE	c5	L2	The number that has to be a range
45	d15	OPS	CHOICE	—	—	What outlives the emergency